Cost Code Creation — Universal Guide

Step-by-Step: How to Create a Cost Code

Step 1 — Create a Cost Code Category (if not exists)

1. Go to **Dashboard** → **Cost Management** → **Categories Tab**
2. Click **"Add Category"**
3. Fill in the fields:
   - **Name** — e.g. `Plumbing` , `Electric` , `Tile Install`
   - **Slug** — lowercase, hyphenated version of name e.g. `tile-install`
   - **Step Number** — which estimator step this category belongs to (1, 2, or 3)
   - **Display Order** — controls sort order in the list
4. Click **Create**

Step 2 — Ensure a Service Exists

1. Go to **Dashboard** → **Web** → **Services**
2. Confirm the service you want to link this cost code to exists
3. If not, create it under the correct **Service Category** and **Project Type**

> Every cost code must be linked to a Service. The Service is the "bathroom type" or "project type" (e.g. Four Piece, Three Piece).

Step 3 — Open the Cost Code Form

1. Go to **Dashboard** → **Cost Management** → **Cost Codes Tab**
2. Click **"Add Cost Code"**

Step 4 — Fill the Service Linking Fields

These are cascading dropdowns — fill them in order:

1. **Project Type** — select the top-level type (e.g. `Bathroom`)
2. **Service Category** — select the mid-level category (e.g. `Bathroom Renovation`)
3. **Service** — select the specific service this cost code belongs to
4. **Category** — select the cost code category (e.g. `Plumbing`, `Electric`)

Step 5 — Fill the Identity Fields

Field	Notes
Code	Unique identifier. Must be unique across ALL cost codes. e.g. `FP-02-07`
Name	Full descriptive title shown in admin
Name Alias (Elies)	Short display name shown in the floating price summary. Optional. If filled, this shows instead of Name.

Step 6 — Add Tips (Optional)

- Click **"+ Add Tip"** to add helpful hints shown to users in the estimator
- Each tip is a single sentence from the user's perspective
- Add as many as needed — one per line
- Example: `"You can relocate your toilet up to 6 feet within the bathroom."`

Step 7 — Add Description (Optional)

- Internal detailed description of the work scope
- Not shown to end users — used for admin reference

Step 8 — Fill Pricing Fields

Field	Notes
Base Price	The builder/labor cost
Markup (%)	Percentage markup. Default is `20`
Client Price	Auto-calculated. Can be manually overridden.

> Client Price auto-calculates as: `Base Price × (1 + Markup / 100)`

Step 9 — Select Unit Type

Unit Type	When to use
`FIXED`	One-time flat cost, no quantity needed
`PER_SQFT`	Price × square footage entered by user
`PER_EACH`	Price × number of units entered by user
`PER_LOT`	Price per lot
`PER_SET`	Price per set
`PER_UPGRADE`	Price per upgrade selection

> If Unit Type is anything other than `FIXED`, also check ✅ **Requires Quantity**

Step 10 — Select Question Type

This controls how the cost code appears and behaves in the estimator. See the **Question Type Reference Card** below.

Step 11 — Fill Step Number & Display Order

Field	Notes
Step Number	Which estimator step this appears in (1, 2, or 3)
Display Order	Sort order within the category. Lower = appears first. Use gaps (10, 20, 30) so you can insert items later.

Step 12 — Nested Question Settings (Only for child questions)

Skip this section if this is a top-level (parent) question.

If this cost code should only appear based on another question's answer:

1. **Parent Question** — select the parent cost code from the dropdown
2. **Show When Parent Value** — the condition that triggers this child (see Nesting Rules below)
3. **Nested Input Type** — how the user interacts with this child

Step 13 — Check the Flags

Checkbox	When to check
✅ **Is Included In Base**	WHITE items that are always part of the base price
✅ **Requires Quantity**	GREEN items — user must enter a number
✅ **Is Optional**	BLUE, YELLOW, GREEN, ORANGE items the user can toggle
✅ **Is Active**	Always check unless it's a placeholder
✅ **Branch Only (Exclude From Export)**	BLUE branch questions with Base Price = `0` — shown in estimator but NOT exported to Buildertrend

Step 14 — Save

Click **Create** (or **Update** if editing).

For **ORANGE** type: after saving, go to **Cost Code Options Tab** and add the dropdown options.

Adding Options for ORANGE Type

After creating an ORANGE cost code:

1. Go to **Cost Management** → **Cost Code Options Tab**
2. Click **"Add Option"**
3. Select the cost code
4. Fill:
   - **Option Name** — display label e.g. `Standard` , `Premium` , `Matte Black`
   - **Price Modifier** — amount added to base price. Can be `0` , positive, or negative.
   - **Display Order** — sort order
   - **Is Default** — ✅ check for the pre-selected option
5. Click **Create**
6. Repeat for each option

> Or: when Question Type = `ORANGE` is selected in the form, an inline options section appears at the bottom — you can add options before saving.

> ⚠️ Always set one option as **Is Default = true**

Question Type Reference Card

Type	Color	User Sees	When to Use
`WHITE`	⬜ White	Fixed line item, no interaction	Always-included scope items
`BLUE`	🔵 Blue	Yes/No toggle switch	Optional add-ons, branch questions
`GREEN`	🟢 Green	Number input field	Quantity-based pricing (sqft, units)

Type	Color	User Sees	When to Use
`ORANGE`	🟠 Orange	Dropdown list	Tiered options (Standard / Premium)
`YELLOW`	🟡 Yellow	Appears after parent = Yes	Follow-up questions after a Yes answer
`PURPLE`	🟣 Purple	Inherits parent's value	Calculated upgrades based on prior answer
`RED`	🔴 Red	Not shown	Placeholder / inactive

Parent-Child Nesting Rules

Scenario	Parent Type	Child Type	Show When Parent Value
Show after parent toggled Yes	`BLUE`	`YELLOW`	`true`
Show after parent toggled No	`BLUE`	`YELLOW`	`false`
Show after parent has any value	`BLUE` / `GREEN`	`PURPLE`	`ANY`
Show after specific dropdown option selected	`ORANGE`	`YELLOW` / `PURPLE`	`{optionId}` — copy from DB
Branch question (no cost, no export)	`BLUE`	any	`true` / `false`

Nested Input Type values:

Value	Meaning
`NONE`	No extra input — child is a simple toggle
`QUANTITY`	Child inherits the parent's quantity value
`DROPDOWN`	Child shows its own dropdown
`CUSTOM_PRICE`	Child allows a custom price entry

> ⚠️ Always create the **parent first**, then create the child and link it.

Pricing Formula

```
Client Price = Base Price × (1 + Markup / 100)

Example:
  Base Price   = $800
  Markup       = 20%
  Client Price = $800 × 1.20 = $960
  Profit       = $960 - $800 = $160  (16.67% margin)
```

`clientPrice` is auto-calculated by the backend when you save. You can manually override it in the form.

For **ORANGE** type:

```
Final Price = Base Price + selected option's Price Modifier

Example:
  Base Price = $3,600
  Option "Standard"  → modifier $0   → total $3,600
  Option "Premium"   → modifier $800 → total $4,400
  Option "Budget"    → modifier -$400 → total $3,200
```

Quick Decision — Which Question Type?

```
Is this work always included, no user choice?
→ WHITE

Does the user turn it on or off?
→ BLUE
  → Also reveals sub-questions? → BLUE (parent) + YELLOW or PURPLE (children)
  → No cost, just branching? → BLUE + Exclude From Export ✅ + Base Price = 0

Does the user enter a number (sqft, units, etc.)?
→ GREEN + Requires Quantity ✅ + Unit Type = PER_SQFT or PER_EACH

Does the user pick from a list?
→ ORANGE + add options after creating

Should this only appear after a Yes answer?
→ YELLOW (child of BLUE, Show When Parent Value = true)

Should this inherit quantity from a previous question?
→ PURPLE (child of GREEN or BLUE, Show When Parent Value = ANY)

Not ready yet / placeholder?
→ RED + Is Active ❌
```

Common Mistakes

Mistake	Fix
Creating a child before the parent	Always create parent first
ORANGE type with no options	Add at least one option after creating
GREEN type without Requires Quantity	Check ✅ Requires Quantity
GREEN type with Unit Type = FIXED	Change to PER_SQFT or PER_EACH
Duplicate code	Each code must be unique — check existing codes first

Mistake	Fix
Branch question with a price	Branch questions should have Base Price = `0`
No default option on ORANGE	Always set one option as Is Default = true
Show When Parent Value left blank on a child	Must fill — `true` , `false` , or `ANY`
Forgetting Exclude From Export on branch questions	Check ✅ Branch Only for questions that only control logic